

Pranav Mishra — Portfolio

AI Engineer and researcher focused on agentic LLM systems, voice AI, and production ML infrastructure. Currently building multi-agent workflows as a founding LLM engineer at [alfred_](#), a Techstars-backed AI assistant startup serving 5,000+ active users. Previously shipped production systems at WheelPrice (consumer-facing CMS for 10–20K daily users) and contribute to applied AI research at the University of Illinois Chicago. Three first-author publications in multi-agent medical reasoning and retrieval-augmented generation, including an accepted paper at IEEE CAI 2026.

Contact: pmishr23@uic.edu · github.com/PranavMishra17 · linkedin.com/in/pranavgamedev · portfolio-pranav-mishra-paranoid.vercel.app

Experience

Founding LLM Engineer — [alfred_](#) (April 2026 - Present)

Building production-grade multi-agent LLM workflows for a consumer AI executive assistant serving 5,000+ active users. Current focus is the evaluation harness for the SMS agent surface: deterministic trace replay, scenario-fixture-driven scoring, and regression detection across verdict types (silent, notify, confirm, clarify, refuse) and tool-calling reliability. Responsibilities span TypeScript backend infrastructure, orchestration reliability, and low-latency context retrieval across real-time pipelines.

AI Engineer — WheelPrice (July 2025 - March 2026)

Built a React, Node.js, and MongoDB content management subsystem from scratch, serving 10,000–20,000 daily users. Implemented Redis caching, server-side rendering, and a dynamic SEO pipeline; reduced API response time from 1.5 seconds to 300 milliseconds through query optimization and connection pooling. Also built an internal analytics dashboard with XGBoost-based churn prediction and event-driven behavior tracking across user sessions.

Research Software Engineer — UIC VARE Lab (February 2024 - Present)

Designed RESTful APIs and a PostgreSQL backend handling over one million records with authentication middleware. Deployed an audio ML inference API at 150-millisecond p95 latency through INT8 quantization. Contributing author on applied AI research in healthcare under the TeamMedAgents project line.

Research and Publications

MetaRAG: A Systematic Framework for Enterprise Knowledge Retrieval

IEEE Conference on Artificial Intelligence (CAI) 2026 — Accepted. First author.

Designed a metadata-enrichment framework for retrieval-augmented generation that uses large language models to generate structured metadata for document chunks, substantially improving semantic representation and retrieval accuracy. The system achieves 82.5 percent precision and a 0.925 Hit Rate at ten through recursive chunking combined with TF-IDF weighted embeddings, and reduces hallucinations by 25 percent through an automated evaluation framework with cross-encoder reranking. Deployed on AWS SageMaker with MLflow for versioning, the system handles 10,000 queries per day with p99 latency under 300 milliseconds.

TeamMedAgents: Enhancing Medical Decision-Making of LLMs Through Teamwork

PAKDD 2026 (under review). First author.

A multi-agent medical reasoning framework that operationalizes the Salas et al. "Big Five" organizational psychology teamwork model into computational agent collaboration. Implements six modular, toggleable teamwork components: team leadership, mutual performance monitoring, team orientation, shared mental models, closed-loop communication, and mutual trust. Evaluated across eight medical benchmarks including MedQA, MedMCQA, PubMedQA, DDXPlus, Path-VQA, and PMC-VQA, with consistent improvements on seven of eight datasets. Controlled ablation studies reveal that optimal teamwork configurations vary by reasoning task complexity and domain.

SLM-TeamMedAgents: Scaling Medical Decision Making with Small Language Models

ICML 2026 submission. First author.

Demonstrates that collaborative Small Language Models in the 500-million to 27-billion parameter range can match the performance of 100-billion-plus parameter models while reducing computational costs by orders of magnitude. Built on Google's Agent Development Kit with Gemma and MedGemma models, the framework supports two to four dynamically recruited specialist agents across a three-round deliberation architecture with trust-weighted voting. Achieves 77.63 percent accuracy across eight benchmarks and $3.1\times$ inference speedup compared to frontier LLMs.

Selected Projects

MockFlow-AI — Real-Time Voice Interview Platform

Python, Flask, LiveKit Agents SDK, Deepgram, OpenAI, Supabase, WebRTC [mockflow-](#)

Production-deployed voice interview platform with finite-state-machine-driven multi-agent orchestration across four interview tracks: introduction, behavioral, technical voice, and technical coding with live Monaco editor. Achieves sub-400-millisecond end-to-end latency through optimized WebSocket connections, five times faster than a polling-based baseline. Implements bring-your-own-keys architecture with encrypted credential storage and per-session ephemeral worker spawning. The FSM design enforces explicit tool-driven stage transitions with minimum-question gates and quality-driven exits, preventing the unstructured drift typical of pure LLM-driven conversations.

alfred_ Execution Decision Layer

Next.js 14, TypeScript, MCP, Cartesia TTS, Vercel

Decision-layer architecture for an agentic AI assistant, designed and prototyped before joining the founding team. Implements five verdict types (silent, notify, confirm, clarify, refuse) over candidate agent actions, with deterministic risk scoring rather than LLM-as-judge for auditability and repeatability. Integrates MCP tool execution, a pending-obligations queue for actions requiring human confirmation, and a bounded undo window for recoverable side effects.

SnakeAI-MLOps — Reinforcement Learning Game Agents in C++

C++17, SFML, LibTorch, PyTorch, TorchScript, Docker, GitHub Actions

End-to-end reinforcement learning system with four algorithms (Q-Learning, DQN with Double and Dueling extensions, PPO, Actor-Critic) trained in Python and deployed to a C++ game engine through TorchScript. The integration layer handles the Python-to-C++ bridge with a single conversion gate, an artifact contract for shape validation, and a `TORCH_AVAILABLE` compile flag for graceful degradation when LibTorch is absent. Median inference latency of approximately 140 microseconds. A 50,000-iteration soak test validated deterministic outputs after resolving a GPU memory corruption bug rooted in tensor lifecycle differences between Python's garbage collector and C++ destruction semantics.

SoulEngine — Agentic NPC Framework

TypeScript, Node.js, Hono, multi-provider LLM (Gemini, OpenAI, Anthropic, Grok), Deepgram, Cartesia, MCP

TypeScript framework for game NPCs with evolving personalities, layered memory, and autonomous agency. Implements a dual-instance Mind architecture per NPC, finite-state-machine memory cycles (Daily Pulse, Weekly Whisper, Persona Shift), MCP tool invocation for world-affecting actions, four voice conversation modes, and YAML-based stateless configuration. Includes a nine-tab web editor with a real-time testing playground.

IVORY — Multi-Modal RAG Avatar Platform for Healthcare Education

Python, FastAPI, Azure Cosmos DB, Azure Speech Services, vector databases

Production-grade retrieval-augmented generation platform with custom conversational avatars for healthcare education. Combines document processing, vector retrieval, text-to-speech generation, avatar rendering, and gesture animation in a unified pipeline with end-to-end latency under 250 milliseconds and near-100 percent retrieval accuracy on benchmark queries.

VoiceLoop — Self-Improving Voice Interview Evaluator

Real-time voice pipeline with stage-bifurcated evaluation architecture

Real-time voice mock-interview platform with a self-improving evaluator loop. Interviewer and evaluator agents operate in a feedback cycle with stage-specific evaluation strategies.

Documents ten production voice-AI failure modes covering turn-taking, latency variance, interruption handling, and transcription drift, with codified resolutions carried forward from prior voice-interview infrastructure work.

Resume-Craft-Pro — AI Resume Optimization Platform

Next.js 14, TypeScript, Gemini API, LaTeX, pdf-lib, Supabase

Production AI utility with dual operating modes: Quick Craft for one-click automated optimization, and Studio Mode for full LaTeX editing alongside an AI chat agent. Implements intelligent keyword extraction, ATS compatibility optimization, parallel Gemini API processing for fan-out generation, and format-preserving component-level edits.

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, C#, Java, Rust

AI and ML: PyTorch, TensorFlow, LangChain, LibTorch, Google Agent Development Kit, LiveKit Agents SDK, Coqui TTS, XTTS-v2, Hugging Face

Backend and Infrastructure: FastAPI, Flask, Node.js, Express, Hono, ASP.NET Core, PostgreSQL, MongoDB, Redis, Supabase, Pinecone, Azure Cosmos DB

Frontend: React, Next.js 14, Tailwind CSS

Cloud and DevOps: AWS (SageMaker, MLflow), Azure (Speech Services, AI Services, Cosmos DB), Docker, Git, CI/CD, Vercel, Railway, Render, Kubernetes

Voice and Audio: Deepgram, OpenAI TTS, Cartesia, Silero VAD, WebRTC, INT8 quantization

Specializations: Multi-agent LLM architecture, retrieval-augmented generation, voice AI pipelines, production LLM deployment, evaluation infrastructure, C++ ML inference, full-stack

Recognition

- **Winner, MIT XR Hackathon 2024** — SnAlder-Cut, a Mixed Reality and generative AI application built on Meta Quest 3 combining spatial mapping, gesture recognition, and AI-driven scene modification.
 - **First Place, HINT 5.0** — Virtual Van Gogh, an interactive NFT museum with Web3 integration and a Unity-to-blockchain data pipeline.
 - **Speaker, INFORMS Analytics+** — Presented the MetaRAG framework to an audience of 700+ analytics professionals.
 - **Accepted, IEEE Conference on Artificial Intelligence 2026** — MetaRAG (first author).
-

Education

Master of Science in Computer Science — University of Illinois Chicago (August 2023 – May 2025). Graduate Assistant. Coursework: Advanced Machine Learning, Applied AI, Advanced NLP, Computer Vision, Virtual Reality, Game Design.

Bachelor of Science in Computer Science and Engineering — Dayananda Sagar College of Engineering, Bangalore (August 2019 – June 2023). Coursework: AI and ML, Software Engineering, Computer Networks, Operating Systems, Data Structures and Algorithms, Database Management Systems.